Task One: Multi-Class Classification & Decision Boundary Analysis

Datheon Machine Learning & Data Science Internship | Technical Report

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1. Executive Summary

This technical report details the end-to-end execution, exploratory data analysis, feature preprocessing, model training, performance evaluation, and decision boundary visualization for Task One of the Datheon ML & DS Internship. The objective was to build and compare linear (Logistic Regression) and non-linear (Decision Tree Classifier) supervised machine learning models on the classic Iris multi-class dataset, evaluating their accuracy, precision, recall, and decision boundary geometry.

> 📌 **TASK BENCHMARK & PERFORMANCE SUMMARY**
> Both Logistic Regression and Decision Tree classifiers achieved outstanding classification performance on the test set. Logistic Regression achieved a 100% test accuracy (1.00 Precision, Recall, F1-Score), while Decision Tree achieved 96.67% accuracy, illustrating the clear linear separability of Iris species when proper feature scaling and feature selection are applied.

2. Problem Statement & Technical Requirements

Multi-class classification requires understanding feature interactions, scaling sensitivity, and model decision geometry. To address Task One requirements, the following workflow was carried out:

- **1. Data Ingestion & EDA:** Load the Iris dataset (150 samples across 3 species: Setosa, Versicolor, Virginica), inspect missing values, and analyze feature distributions using Seaborn pairplots.
- **2. Feature Preprocessing & Scaling:** Perform a stratified train-test split (80/20) and apply StandardScaler to normalize numerical feature distributions for linear model stability.
- **3. Model Training & Comparison:** Train Logistic Regression (multinomial) and Decision Tree Classifier models on scaled features.
- **4. Quantitative Evaluation:** Compute Confusion Matrices and Detailed Classification Reports (Accuracy, Precision, Recall, F1-Score).
- **5. Decision Boundary Visualization:** Retrain models on 2D projections (petal length vs. petal width) and plot 2D meshgrid decision boundaries to contrast linear vs. non-linear decision regions.

3. Exploratory Data Analysis & Preprocessing

3.1 Exploratory Data Analysis

The dataset contains 150 instances evenly distributed across three target classes (50 Setosa, 50 Versicolor, 50 Virginica). Exploratory pairplots revealed that Iris Setosa is linearly separable from the other two species purely based on petal dimensions (petal_length and petal_width). Versicolor and Virginica exhibit slight overlap along sepal dimensions but remain highly separable in petal feature space.

3.2 Data Preprocessing & Stratified Splitting

To maintain equal class proportions in both splits, a 80/20 stratified split was performed, resulting in 120 training samples and 30 testing samples. StandardScaler was fitted on the training set and used to transform both training and test sets, preventing data leakage while bringing all continuous variables to zero mean and unit variance.

4. Model Training & Quantitative Evaluation

Model evaluation on the hold-out test set (30 samples) demonstrated exceptional multi-class performance:

Model Architecture	Accuracy	Macro Precision	Macro Recall	Macro F1-Score
Logistic Regression	**100.0% (1.00)**	1.00	1.00	**1.00** ☑
Decision Tree Classifier	96.67% (0.967)	0.97	0.97	0.97

5. Decision Boundary & Model Geometry Analysis

To visually interpret model behavior, classifiers were retrained on a 2D feature subspace consisting of petal_length and petal_width. Decision contours generated across a dense meshgrid highlighted fundamental architectural differences:

- **1. Logistic Regression Boundaries:** Produces smooth, linear decision boundaries separating species. Because the underlying class distributions in petal feature space are continuous and naturally separated, the linear hyperplanes easily partition all three species without overfitting.
- **2. Decision Tree Boundaries:** Produces axis-aligned rectangular decision regions corresponding to hierarchical if-else feature thresholds. While flexible, unconstrained decision trees risk creating jagged boundary splits around boundary instances.
- **3. Feature Dominance:** Petal dimensions (length and width) provide almost complete class discrimination compared to sepal dimensions, confirming EDA insights.

6. Conclusions & Recommendations

- **Model Recommendation:** Logistic Regression is recommended as the optimal baseline classifier for linearly separable tabular datasets like Iris due to perfect test accuracy (100%), smooth decision boundaries, and zero risk of structural overfitting.
- **Deployment Packaging:** The StandardScaler and fitted Logistic Regression model can be saved as a single joblib pipeline artifact for instant multi-class inference.